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**Author for correspondence:**
Chloé Julie Loïs Fourreau
e-mail: chloisf@gmail.com

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# The Trojan seahorse: citizen science pictures of a seahorse harbour insights into the distribution and behaviour of a long-overlooked polychaete worm

Chloé Julie Loïs Fourreau[1], Laura Macrina[3], Jue Alef A. Lalas[1], Ai Takahata[1], Tatsuki Koido[4] and James Davis Reimer[1,2]

[1]Molecular Invertebrate Systematics and Ecology Laboratory, Graduate School of Engineering and Science, and [2]Tropical Biosphere Research Center, University of the Ryukyus, 1 Senbaru, Nishihara, Okinawa 903-0123, Japan
[3]Marine Science Program, Biological and Environmental Science and Engineering Division, King Abdullah University of Science and Technology (KAUST), Thuwal 23955-6900, Saudi Arabia
[4]Kuroshio Biological Research Foundation, 560 Nishidomari, Otsuki, Kochi 788-0333, Japan

CJLF, 0000-0002-0062-2876; LM, 0000-0003-1863-0134; JAAL, 0000-0003-4254-0569; TK, 0000-0003-1004-5686; JDR, 0000-0003-0453-8804

Symbiotic marine invertebrates can be small, hidden or difficult to find, hampering the understanding of their distribution and ecological roles. *Haplosyllis anthogorgicola* is a polychaete inhabiting the gorgonian *Anthogorgia bocki*, where it lives in high densities within the host's coenenchyme and occupies burrows formed by host tissue near coral polyps. This study provides the first records of *H. anthogorgicola* since its description in 1956, from colonies of Anthogorgiidae in southern Japan. We observed that host gorgonians were also inhabited by the pygmy seahorse *Hippocampus bargibanti*, a popular species to observe and photograph among SCUBA divers. Therefore, we examined photographic records of *H. bargibanti* available on the citizen science website iNaturalist and screened for structures associated with infestation by *H. anthogorgicola* to gather information on this elusive species. Our analyses confirmed that this polychaete and/or similar species are widespread in the central Indo-Pacific region. In addition, we observed some polychaete behaviours, raising questions about the nature of the relationships between *H. anthogorgicola*, its gorgonian hosts and the pygmy seahorse. Our study demonstrates that citizen science can contribute to our knowledge not only on the distribution and behaviour of well known and charismatic species but also inadvertently on overlooked and neglected taxa.

## 1. Introduction

Symbiotic relationships among marine invertebrates are a key component of diversity, structured by different patterns of host and inhabitant specificities [1–4], with species exhibiting a wide and fascinating range of morphological adaptations [5–7], feeding strategies [8,9], reproductive strategies [10,11] and other behavioural traits [12,13]. However, in some of these associations, symbionts can have a small size [14], can bear colorations mimicking that of their host [15–18] or live hidden inside it [19–21], making these relationships intrinsically difficult to study [11,14].

Among such symbiotic organisms, *Haplosyllis* Langerhans, 1879 is a diverse genus of polychaetes belonging to the family Syllidae Grube, 1850 [22–25] and including many species living in close association with sponges [22,26], with the nature of their relationships varying from predatory, parasitic, and mutualistic [11,26]. Uniquely, three species have been documented living

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in an obligatory association with gorgonian hosts: *Haplosyllis anthogorgicola* Utinomi, 1956 from southern Japan, *Haplosyllis chamaeleon* Laubier, 1960, from the Mediterranean Sea and Atlantic coasts of the Iberian Peninsula and *Haplosyllis villogorgicola* Martin, Núñez, Riera & Gil, 2002 from the Canary Islands. Two of the species are documented from a single host, namely *Anthogorgia bocki* Aurivillius, 1931 for *H. anthogorgicola* and *Villogorgia bebrycoides* (von Koch, 1887) for *H. villogorgicola,* while *H. chamaeleon* inhabits *Paramuricea clavata* (Risso, 1827) in the Mediterranean Sea [27–29] and *Paramuricea grayi* (Johnson, 1861) in the Atlantic [22]. Each of these species show distinct strategies and adaptations to their hosts, but only *H. anthogorgicola* lives in high densities under the host's coenenchyme (tissue surrounding polyps in octocorals). Its infestation rate may reach up to 15 worms per $cm^2$ of gorgonian tissue [29], which is comparable to levels of tropical sponge-dwelling species of *Haplosyllis* [26,30]. In contrast, *H. chamaeleon* and *H. villogorgicola* have much lower infestation intensities, with up to two specimens of *H. chamaeleon* found per metre of gorgonian [31]. In addition, *H. anthogorgicola* induces modifications of the host's tissue that include galleries as well as tube-shaped outgrowths made from host tissue and located in proximity to its polyps, often also oriented towards them [27,29]. Worms were observed occupying and protruding out of these burrow-like outgrowths [27]. Considering this position and the morphology of *H. anthogorgicola,* which has long anterior appendages (namely antennae, tentacular cirri and the first dorsal cirri), Martin *et al*. [29] suggested that the worms could be coral cleaners or kleptoparasites, using the outgrowth position to steal food from the coral polyps. The high infestation intensity along with the putative behaviour of *H. anthogorgicola* raise questions about the exact nature of its relationship with their gorgonian hosts, which are able to maintain themselves while supporting such a large population of worms.

Despite the singularity of *H. anthogorgicola* in life habits and characteristic modifications of host tissues, the species has not been recorded since its original description by Utinomi [27]. In this study, we provide new records in southern Japan, thus extending its distribution further south from the type locality. At these new locations, we observed that the galleries and outgrowths induced by the species were visible underwater, and that another symbiont of gorgonians, the pygmy seahorse, *Hippocampus bargibanti* Whitley, 1970, was present. *Hippocampus bargibanti* is a popular species to observe among recreational divers [32,33], who frequently photograph it and upload images onto the internet. Recently, the use of citizen science platforms has brought insights into the distribution of species, allowing the documentation of rare species [34] and the description of new species [35], as well as the observation of unique and important behaviours in marine species [36], which would have been difficult to report solely relying on scientific surveys. Owing to the relatively small size of the seahorse, many contributors take close-up photographs, which concomitantly show portions of the gorgonian host, making structures associated with *H. anthogorgicola* infestations observable. Therefore, we examined these images to assess the presence of *H. anthogorgicola,* infer its distribution and describe its behaviour.

## 2. Material and methods

### (a) Sampling and fixation

Branches of gorgonian octocorals (approximately 5 cm long) were collected with pliers by SCUBA diving at 25 m depth in Sukumo, Kochi, Japan from a colony in February 2023, at 20 m in June 2023 and January 2024 at Cape Zanpa, Okinawa-jima, Japan, and at 26 m in July 2023, Kashiwajima Bay, Kochi, Japan. Octocoral colonies were photographed underwater using an Olympus Tough TG-6 waterproof camera in PT-059 underwater housing. For the colony in Sukumo, polychaete worms were noticed on the samples and individually separated from gorgonian branches and fixed in 99% ethanol. For the other colonies, live gorgonian branches were kept in 50 ml Falcon tubes filled with seawater for a few hours after sampling, after which most polychaete worms had exited the colony, allowing us to sort the worms and fix them and the gorgonian branches separately in 80 or 99% ethanol.

### (b) Morphological analyses

Worms were examined using a Nikon Eclipse 80i microscope and a Hitachi Miniscope TM4000PlusII scanning electronic microscope (SEM). Additional photographs of the collected polychaetes were taken using a Sony Alpha camera mounted on a Leica S8AP0 stereoscope through a Micronet lens adapter. Identification of the worms was based on the descriptions and re-descriptions of gorgonian-dwelling *Haplosyllis* [29]. To identify the octocoral colonies from Kashiwajima, Kochi and Zanpa, Okinawa, sclerites were prepared for SEM analyses separately by bleaching tissue of the coenenchyme, the polyp wall and polyp tentacles. Samples were then washed with distilled water and mounted on SEM stubs. Sclerites of the colony from Zanpa, Okinawa were dried and platinum-coated with an automatic fine coater (JEOL model JFC-1600) and imaged using a JEOL JSM-6510 SEM, whereas those of the Kashiwajima, Kochi colony were visualized on a Hitachi Miniscope TM4000PlusII SEM.

### (c) Retrieval of information on *Haplosyllis anthogorgicola* from iNaturalist records of *Hippocampus bargibanti*

In order to extract potential information on the behaviour and distribution of *H. anthogorgicola,* we examined photograph records of *H. bargibanti* on the iNaturalist website (https://www.inaturalist.org/). Metadata of 651 research-grade photographic records of *H. bargibanti* were downloaded on 9 July 2023 from the citizen science website iNaturalist through the observation page (https://www.inaturalist.org/observations/), with the entry 'pygmy seahorse' and the sole filter 'verified' activated.

To assess the presence of *H. anthogorgicola* in each record and infer the worm's distribution, photographs were examined for the presence/absence of burrows surrounding polyps (thereafter 'polyp burrows'), burrows along the gorgonian branches

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('branch burrows'), galleries and visible worms over two rounds of review of the dataset. When no structure was visible, the quality of the photograph was also recorded, particularly if it was blurry, dark or taken too far away to reasonably distinguish details of structures. The GPS coordinates of photographs for which at least one worm structure could be observed were extracted and mapped along with specimen-based records using RStudio v. 1.4 [37], ggmap [38] and maptools [39] packages. Dubious coordinates pointing to land more than 50 km distant from the nearest coastline were removed. To confirm the host range, a third round of examination of the dataset was performed to identify the gorgonian colonies in the iNaturalist pictures. These were identified to family level based on original descriptions, branching patterns and polyp shape and arrangement [40–42].

To gain insights into the behaviour of *H. anthogorgicola* and its relationships with gorgonian hosts and *H. bargibanti*, for each observable worm in the iNaturalist photographs, we recorded the following five behavioural categories: (1) Location of worm, as the structure on or in which most of the worm's body was seen; this included the levels 'In polyp burrow', 'In branch burrow' and 'On pygmy seahorse'; (2) Degree of exit from gorgonian host, including the following five levels: 'Protruding appendages' (only the cirri or antennae were visible), 'Protruding head' (a part of the palps or prostomium was visible), 'Protruding anterior body' (the proventricle, a characteristic muscular region in the gut of Syllidae, was visible), 'Visible body' (the whole body was outside of the gorgonian host) and 'Protruding posterior body' (only the posterior end of the worm was visible from a burrow); (3) Orientation of worm; the first object directly situated in the direction of the anterior part of the worm was noted as the orientation, including the 11 levels: 'Closed polyp', 'Water column', 'Pygmy seahorse', 'Branch', 'Undetermined', 'Close worm', 'Open polyp', 'Branch burrow', 'Debris', 'Inner gallery' and 'Polyp burrow'; (4) Distance approximation between worm and pygmy seahorse; as the photographs had no scale, we could not reliably estimate a real distance and reported a qualitative approximation of the distance separating each worm and the nearest pygmy seahorse in each picture relative to the possibility of contact, based on the size and potential movement of the worms only: 'In contact', 'Contact possible', 'Close but too far for contact' and 'Distant'; (5) Orientation of pygmy seahorse; this category described the orientation of the closest seahorse relative to each worm, including eight levels: 'Water column', 'Distant branch to worm', 'Distant polyp to worm', 'Distant branch burrow', 'Close polyp to worm', 'Close branch to worm', 'Different worm' (the seahorse was directed towards a different worm from the one being recorded) and 'This worm' (the seahorse was directed towards the worm being recorded). To detect patterns between the different behavioural categories, a multiple correspondence analysis was performed on the resulting dataset. The analysis was conducted in RStudio v. 1.4 [37] using packages FactoMineR [43], factoextra [44] and ggplot2 [45]. This analysis was redone excluding levels of categories that were considered as outliers, that skewed the multivariate representation (namely Location of worm: 'On pygmy seahorse', 'Undetermined'; Orientation of worm: 'Undetermined', 'Polyp burrow', 'Inner gallery'; Orientation of pygmy seahorse: 'Branch burrow'). Using R package stats [46], we also tested the chi-squared distribution of the pairs of categories to evaluate their association. Subsequently, *post hoc* tests were performed with Bonferonni correction to understand which levels of categories were associated, using the package chisq.posthoc.test [47].

## 3. Results

### (a) Morphology of the polychaetes

The highly abundant polychaetes found inside the gorgonians from Zanpa, Okinawa and from Kashiwajima and Sukumo in Kochi matched the re-description of *H. anthogorgicola* by Martin *et al*. [29]. The worms had articulated cirri and antennae particularly long anteriorly, with the dorsal tentacular cirri sometimes extending as far or farther the median antenna. Dorsal cirri had lengths alternating between long (L) and short (S) in the following pattern from the dorsal cirri at first chaetiger: L-S-S-L-S-L-S-S-S and continuing short until the end of the body. All chaetae were bidentate and the anterior ones had an incision on the main fang and a damp triangular protrusion on the opposite side. Chaetae changed in shape most noticeably at the 6–8th chaetiger, from which the incision on the top of the main fang became dampened and disappeared. The main fang became wider and longer in the following mid-body and posterior chaetae, and protrusion on the shaft dampened (figure 1*f*). In addition, stolons were cephalous and dicerous, with a pair of lateral antennae and two pairs of very close red eyes in a trapezoidal arrangement, with the anteriormost pair having larger, more distantly separated eyes than the posterior ones (figure 1*b*). Antennae on the top of the prostomium were triangular with rounded tips. Stolons had 13–15 chaetigers and showed a non-regionalized digestive tract. *In vivo*, female stolons carried large pink eggs (figure 1*b*), while male stolons carried sperm and had a cream-whitish colour (figure 1*c*). In addition to capillary natatory chaetae, stolons had 1 or 2 chaetae, similar in shape to the mid-body and posterior ones in the atokous form (figure 1*d*).

### (b) Morphology of the gorgonians

The three gorgonian colonies collected matched the diagnosis of Anthogorgiidae McFadden, van Ofwegen & Quattrini, 2024, a family of octocorals characterized by having monomorphic, non-retractile polyps, with sclerites of the polyps arranged in chevrons, in the shape of blunt spindles bearing large tubercles, sclerites of the tentacles consisting of small rods and sclerites of the coenenchyme similar to those of the polyps, with the addition of capstans. Anthogorgiidae includes genera *Muricella* Verrill, 1869 and *Anthogorgia* Verrill, 1868, which are distinguished by the height and shape of non-retractile polyps, the size of spindles in the coenenchyme and the degree of the anastomosis of branches [42,48–50]. The gorgonian colony from Sukumo, Kochi was identified as *Anthogorgia* sp. in a previous publication [50]. The gorgonian colony from Kashiwajima closely resembled that from Sukumo, having uniform size of sclerites from coenenchyme surface, tall polyps and few net-like branch fusions. In contrast,

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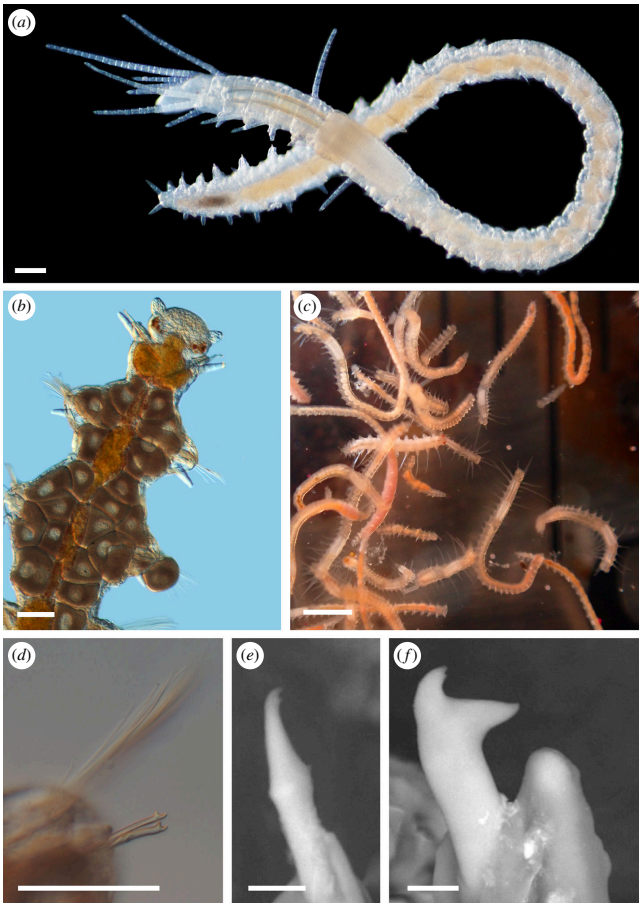


**Figure 1.** *Haplosyllis anthogorgicola.* (*a*) Live atokous worm, (*b*) live anterior part of a female stolon, (*c*) live worms including male stolons, atokous worms and maturing female stolons with oocytes, (*d*) natatory and simple hooked chaetae of a female stolon, scanning electron microscope image of chaetae, (*e*) anterior chaeta of an atokous worm, (*f*) posterior chaeta of atokous worm. Scales: (*a*,*b*,*d*): 0.1 mm, (*c*): 1 mm, (*e*,*f*): 5 µm.

the gorgonians from the colony from Zanpa, Okinawa contained surface sclerites in coenenchyme that were clearly larger in size than those elsewhere, and the polyps were short and dome-shaped, with some branch fusions (figure 2*b*), characters that better correspond to *Muricella.* In addition, the sclerites of the polyps and coenenchyme, having large tubercles, bore closer resemblance to *Muricella* spp. reported by Reijnen *et al*. [32] than to the colonies from Kochi [50]. However, owing to the taxonomic uncertainty surrounding Anthogorgiidae [42], any genus-level identifications must be considered tentative, and hence we presently assign our specimens to Anthogorgiidae.

## (c) Observations of *Hippocampus bargibanti* and structures of *Haplosyllis anthogorgicola* on gorgonians

*Hippocampus bargibanti* was present on colonies of Anthogorgiidae at Kashiwajima, Kochi (figure 3*a*) and at Zanpa, Okinawa (figure 3*i*), in addition to on a previously reported colony at Sukumo, Kochi [50]. The seahorses were attached to gorgonian branches and oriented towards the colony in all of our observations.

On all directly observed gorgonian colonies, galleries made of the host tissue were visible from outside the colonies, on the stalk and on branches of varying sizes (figure 3), although they appeared to be absent or more difficult to observe on smaller branches where polyps were densely distributed (figure 3). Galleries appeared to follow the branch axes, connecting polyps throughout their paths, and further galleries appeared to connect parallel galleries (figure 3*a–e,g,h*). Burrows of *H. anthogorgicola* showed the same colour as the host tissue but stood out by having a small area with no sclerites around their opening (figure 3*c,f* ). Burrows on the colony from Zanpa, Okinawa appeared as protrusions of the coenenchyme that were slightly more elongated than the ones from Kochi (figure 3). Burrows on the branches were distributed along the path of galleries (figure 3*a,b,d,e,g,h*), and burrows near polyps were also observed on those branches on which galleries were visible. Burrows near polyps were abundant in smaller branches (figure 3*c,f*) but were harder to observe underwater, probably owing to the density of the polyps and sclerites in such ramifications.

## (d) Distribution of *Haplosyllis anthogorgicola* based on iNaturalist photographic records

Ninety-one records did not show the presence of the worms despite matching detection requirements (electronic supplementary material, table S1). At least one structure indicating the presence of *H. anthogorgicola* or a similar species was observable in the remaining 489 photographs (75.1% of photographs). However, infestation by worms was considered unclear in 71 out of 651 records owing to some photographs not showing gorgonians, being taken too far from the gorgonian colony, or being

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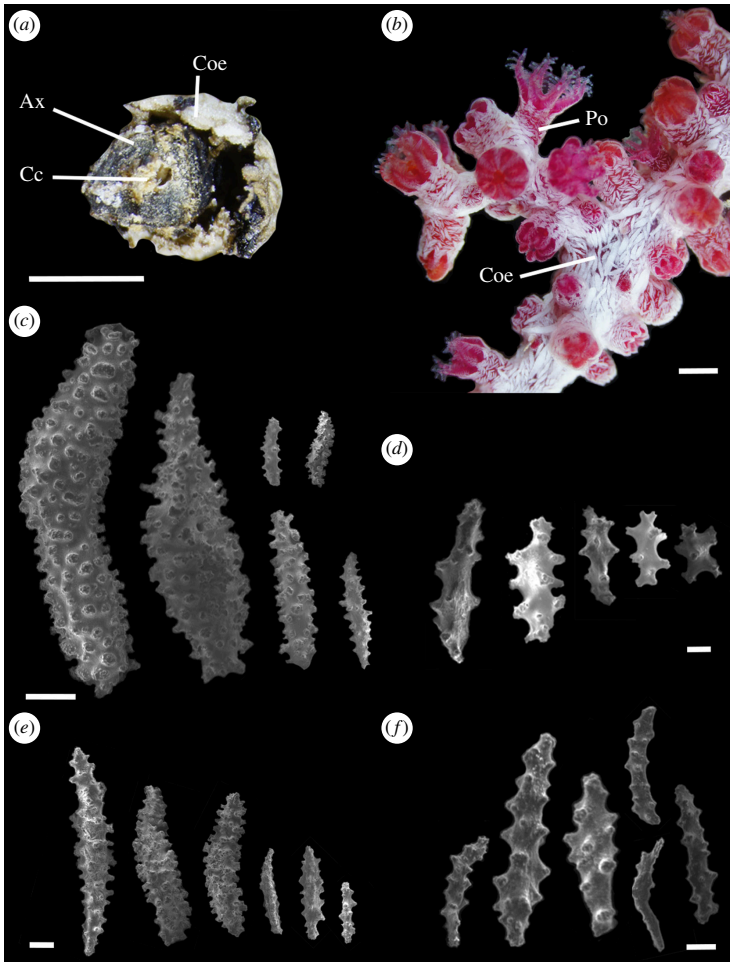


**Figure 2.** Morphological features of the Anthogorgiidae colony from Zanpa, Okinawa. Stereomicroscope micrographs of (*a*) cross-section of branch axis (Ax, axis; Coe, coenenchyme; CC, central core), and (*b*) branch with closed and open polyps visible (Coe, coenenchyme; Po, polyp). Sclerites observed by scanning electron microsope: (*c*) spindles from the outer layer of coenenchyme, (*d*) capstans from the inner layer of coenenchyme, (*e*) spindles in the polyp walls, and (*f*) small rods in tentacles. Scales: (*a*,*b*): 1 mm, (*c*): 100 µm, (*d*,*f*): 20 µm, (*e*): 50 µm.

of unsuitable quality to allow observation of details on the gorgonian. Excluding the 71 poor quality photographic records, the percentage of detected infected gorgonians rose to 84.1% (=489/580). The most frequently detected structures were polyp burrows, followed by galleries, branch burrows and visible worms (electronic supplementary material, figure S1). As for the geographical distribution, GPS coordinates associated with records displaying signs of *H. anthogorgicola* indicated potential distribution in Japan, Taiwan, Malaysia, the Philippines, Indonesia, East Timor, Papua New Guinea, Australia and New Caledonia (figure 4). The oldest photograph showing structures of the worm was taken in 2001 and uploaded in 2015 (ID 1516840), whereas the first picture that was shared on iNaturalist was uploaded in 2012 and taken in 2008 (ID 163192).

## (e) Behaviour of *Haplosyllis anthogorgicola* based on iNaturalist photograph records

Worm behaviour was assessed from 448 worms observed in 105 photographic records (figure 5; electronic supplementary material, table S2 and table S3). 'Location of worm' and 'Orientation of worm', 'Degree of exit from gorgonian host' and 'Orientation of worm', 'Degree of exit from gorgonian host' and 'Distance approximation between worm and pygmy seahorse', 'Orientation of worm' and 'Distance approximation between worm and pygmy seahorse' showed significantly strong relationships (chi-squared, $p < 0.001$), while 'Orientation of pygmy seahorse' and 'Distance approximation between worm and pygmy seahorse' were also significantly associated (table 1). Multiple pairs of levels were either positively or negatively associated among the comparisons between categories (electronic supplementary material, figure S4). The first multiple correspondence analysis (electronic supplementary material, figure S2) showed strong clustering related to the orientation of the worm, their location and degree of exit from gorgonian hosts. The degree of exit from gorgonian host: 'Visible worms' was associated with the location of worm; 'On the pygmy seahorse' while Orientation of worm: 'Inner gallery' was exclusively associated with the degree of exit from gorgonian host: 'Protruding posterior body', as also revealed by the chi-squared *post hoc* tests (electronic supplementary material, figure S4). The relationship between other levels was difficult to see due to these associations. Figure 6 shows the multiple correspondence analysis performed after removing these outliers. Most worms were observed with only their anterior appendages protruding out of the gorgonian host ($n$ = 318, electronic supplementary material, table S3), which appeared closely associated with worms being located in the polyp burrows and being oriented towards either closed polyps or the water column (figure 6; electronic supplementary material, figure S3 and S4). The central position of 'Protruding anterior body', surrounded by many worm orientations (figure 6), indicated a high diversity of interactions in this case, including

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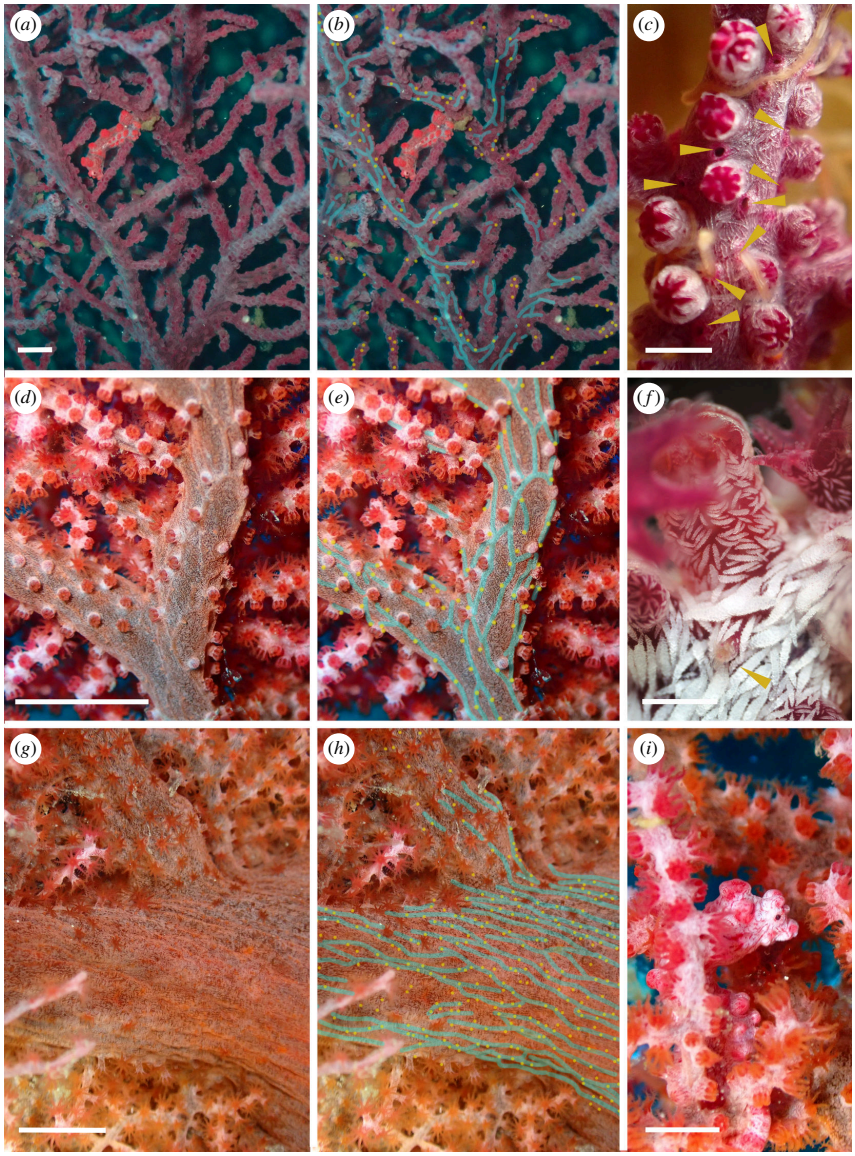


**Figure 3.** Photographs of an anthogorgiid host from Kashiwajima, Kochi: (*a*) pygmy seahorse, worm galleries and burrows; (*c*) close up on a branch, with visible burrows and protruding worms. Photographs of an anthogorgiid host from Zanpa, Okinawa: (*d*) small branch with galleries and burrows visible, (*f*) burrow inhabited by a worm at the base of a coral polyp, (*g*) base of the colony with galleries and burrows visible, (*i*) pygmy seahorse on the colony. (*b*,*e*,*h*) The same photographs that the ones preceding them, with galleries and burrows highlighted in blue and yellow, respectively. Yellow arrows in (*c*,*f*) indicate burrow openings. Scales: (*a*,*d*,*g*) 1 cm, (*c*,*f*,*i*) 1 mm.

worms being oriented towards the pygmy seahorse, debris, closed polyp, branch, branch burrow and the water column, which did not show any significant association (electronic supplementary material, figure S4). However, worms oriented at the pygmy seahorse were particularly associated with being either in direct contact, or at a distance allowing contact with the pygmy seahorse (figure 6; electronic supplementary material, figure S3 and S4).

## 4. Discussion

### (a) Distribution of *Haplosyllis anthogorgicola*

The polychaetes from Kashiwajima, Sukumo and Zanpa corresponded to the morphology reported in the original description [27] and subsequent re-description [29] of *H. anthogorgicola*. Thus, to the best of our knowledge, they represent the first records of the species since its original description, while extending the species distribution towards more southern parts of Japan. Furthermore, the observation of structures attributable to infestation by *H. anthogorgicola* (or a similar species), namely polyp burrows [27], galleries [29] and branch burrows, in iNaturalist photographs (figure 4) suggested a wide distribution in the central Indo-Pacific, a geographical pattern closely matching the distribution of *H. bargibanti* [51]. Gorgonian-associated *Haplosyllis* populations might be more widely distributed than we hypothesized here, and may have not been detected elsewhere owing to the inherent limitations of the dataset being based on images of the seahorse. This agrees with the report of *Haplosyllis spongicola* (Grube, 1855) [52] from the Indian Ocean, on the gorgonian *Muricella ramosa* Thomson & Henderson, 1905. This report may correspond to *H. anthogorgicola* or a similar species, but certainly not to the Mediterranean *H. spongicola*

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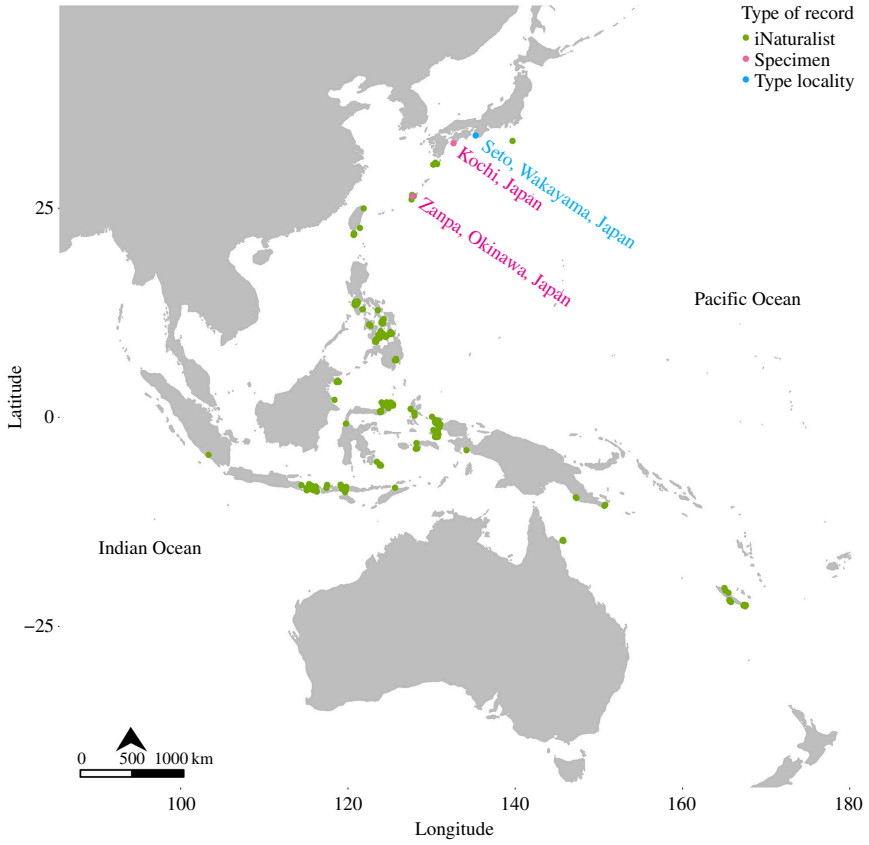


**Figure 4.** Map of the distribution of *Haplosyllis anthogorgicola* as inferred from specimen records and from the iNaturalist dataset.

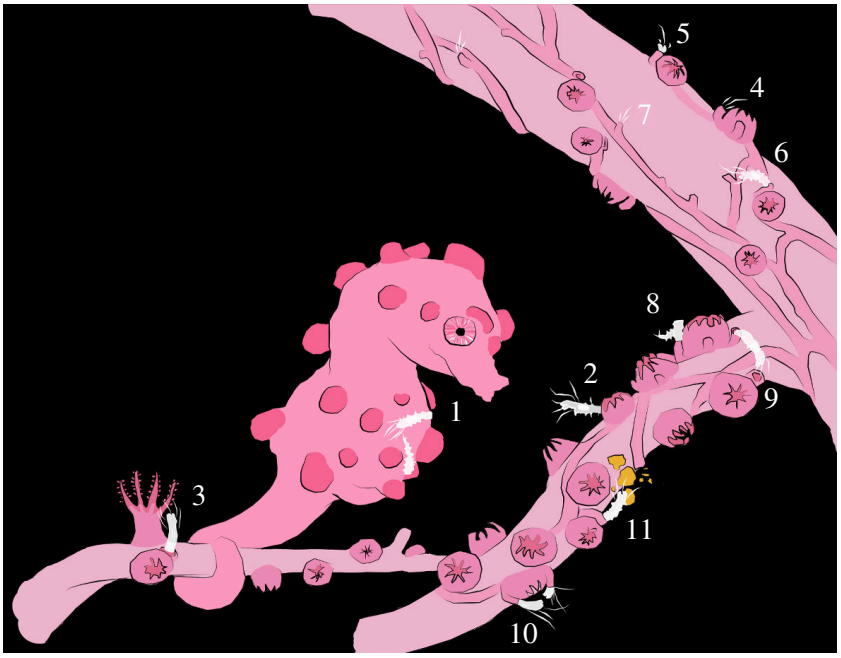


**Figure 5.** Schematic summary of the behaviours of *Haplosyllis anthogorgicola* observed in the iNaturalist dataset. Each worm is indicated by a number. Behaviours of the worms include the location of worm, degree of exit from gorgonian host and orientation of the worm, as well as the distance approximation between worm and pygmy seahorse. (1) Worm located on pygmy seahorse, with visible body, oriented towards the pygmy seahorse, in contact. (2) Worm located in polyp burrow, with protruding anterior body, oriented towards the pygmy seahorse, close but too far for contact. (3) Worm located in polyp burrow, oriented towards open polyp, with protruding anterior body, contact possible. (4) Worm located in polyp burrow, with protruding appendages, oriented towards closed polyp, distant. (5) Worm located in polyp burrow, with protruding head, oriented towards water column, distant. (6) Worm located in polyp burrow, with protruding anterior body, oriented towards branch burrow, distant. (7) Worm located in branch burrow, with protruding appendages, oriented towards branch, distant. (8) Worm located in polyp burrow, with protruding posterior body, oriented towards inner gallery, close but too far for contact. (9) Worm located in polyp burrow, with protruding anterior body, oriented towards polyp burrow, close but too far for contact. (10) Worm located in a polyp burrow, with protruding anterior body, oriented towards close worm, close but too far for contact. (11) Worm located in polyp burrow, with protruding anterior body, oriented towards debris, close but too far for contact. Behaviours and their corresponding illustration number and occurrences are listed in electronic supplementary material, table S3.

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**Table 1.** Results of the chi-squared distribution tests for the different pairs of behavioural categories of *Haplosyllis anthogorgicola*, showing the probability values (*p*). *$p < 0.05$; **$p < 0.01$; ***$p < 0.001$.

| pairs of categories tested for chi-squared distribution | *p*-value |
|---|---|
| Location of worm–Degree of exit from gorgonian host | *** |
| Location of worm–Orientation of worm | *** |
| Location of worm–Orientation of pygmy seahorse | 0.32 |
| Location of worm–Distance approximation between worm and pygmy seahorse | *** |
| Degree of exit from gorgonian host–Orientation of worm | *** |
| Degree of exit from gorgonian host–Orientation of pygmy seahorse | * |
| Degree of exit from gorgonian host–Distance approximation between worm and pygmy seahorse | *** |
| Orientation of worm–Orientation of pygmy seahorse | 0.06 |
| Orientation of worm–Distance approximation between worm and pygmy seahorse | *** |
| Orientation of pygmy seahorse–Distance approximation between worm and pygmy seahorse | *** |

[11]. Structures observed in iNaturalist photographs unequivocally resemble the structures described by Utinomi [27] and re-described by Martin *et al*. [29], as well as directly observed in the present study (figure 3). However, specimens of the symbiotic worms from all these other locations are required to confirm whether such records can certainly be assigned to *H. anthogorgicola*, or if they are of a different, closely related species of *Haplosyllis* exhibiting similar habits.

## (b) Host specificity and infestation

The finding of *H. anthogorgicola* on several Anthogorgiidae with different morphologies suggests that this polychaete species inhabits a larger diversity of hosts than previously known. In line with this, different morphologies of the hosts were observed in our iNaturalist dataset, such as differences in colour, and branching patterns and degrees of anastomosis, and shape of the polyps. However, owing to the taxonomic uncertainty surrounding the genera *Muricella* and *Anthogorgia*, which are morphologically highly similar, genetically very close and poorly studied [42], it is currently difficult to identify the hosts to genus or species level. Nevertheless, most of the gorgonian colonies hosting *H. bargibanti* observed in our iNaturalist dataset appeared to be infested with *H. anthogorgicola*, as indications of the worm's presence were found on 75% of the photographic records investigated. However, the poor quality of some photographs [53], combined with the difficulty to distinguish the worm-related structures on the smaller branches or when polyps were open, even on our own photographs (figure 3*d,e,i*), suggest that the worms' presence has been underestimated in our analyses. This together suggests that *H. anthogorgicola* may be ubiquitous on Anthogorgiidae.

## (c) Behaviour of *Haplosyllis anthogorgicola*

Burrows close to the polyps (figure 3) were visible in living gorgonians even underwater, as they were drawn by Utinomi [27] and later photographed by Martin *et al*. [29]. Worm galleries were also visible on the external parts of the gorgonian colonies in the wild, despite being originally described as visible only in the inner side of the gorgonian tissue [27,29], probably because these authors observed smaller, preserved branches. Galleries were especially visible on larger branches and appeared to form a network connecting polyps (figure 3*a,b*), highlighting that access to polyps is important for *H. anthogorgicola*. Worms occupying polyp burrows were strongly associated with orientation towards a closed polyp (worm 4 in figure 5) or towards the open water column (worm 5 in figure 5), and in most of the observations, only their appendages were protruding (figure 6 and electronic supplementary material, table S3). This suggests that worms occupy polyp burrows to detect movements of the polyp or disturbances in the water column. These observations agree with Martin *et al*. [29], who suggested that the long length of appendages and marked change of chaetal shape in the anterior region could be an adaptation for protruding out of burrows to steal food from polyps. However, and although worms feeding on particles caught by coral polyps may be an interaction difficult to capture, we rarely observed worms interacting with fully or partially open polyps. In fact, the only times that worms were observed in contact with an open polyp (photograph ID 1616784), they were also located at close distance to a pygmy seahorse. Rather than the worms directly aiming to interact with the polyp, they may have been agitated by the presence of the pygmy seahorse. Other explanations, such as cleaning the host tissues and guarding the polyps (see below), are also possible. Our observations suggest that worms strongly rely on associated structures other than the polyps to move and interact with the external environment, as many burrows of *H. anthogorgicola* were situated along the gallery paths, distant from polyps (figure 3*b,d*). Their abundance and placement suggest that worms use them for other uses than access to the polyps, such as detection of surrounding movement, capturing of food and exiting galleries. In addition, some worms were seen interacting with debris trapped in the mucus on the colony (iNaturalist ID 38953421), suggesting they may behave as cleaners, potentially feeding on organic matter in such debris.

Most of the orientations of the worms imply protruding the anterior body outside of the host, suggesting that most of their interactions with the exterior are carried out while only partially exiting the host—a position likely allowing them to quickly hide in the face of a threat. This is also in line with the general abundance of burrows, granting them ample access to the colony's

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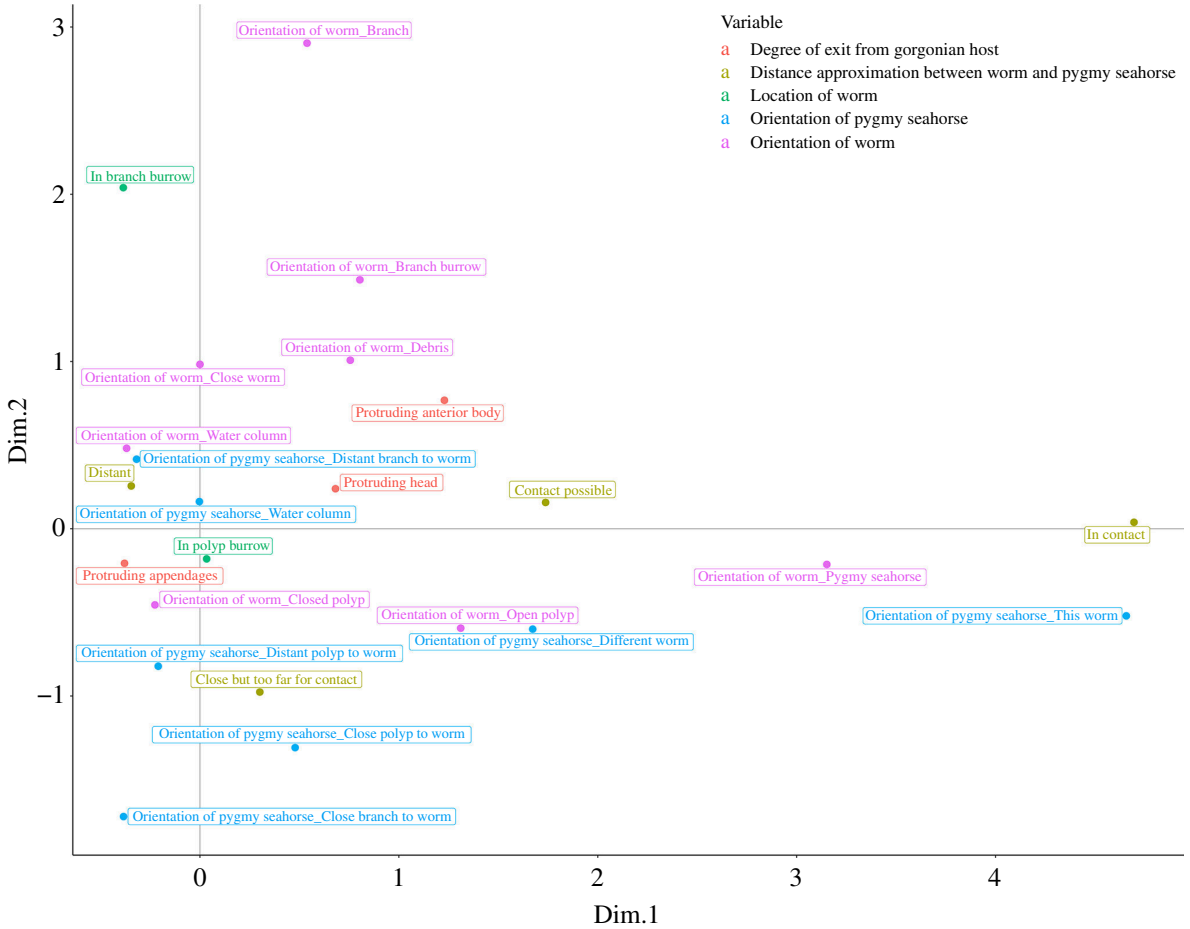


**Figure 6.** Multiple correspondence analysis plot of variable categories representing the behaviour of *Haplosyllis anthogorgicola* with respect to the location of the worm, its degree of exit from the gorgonian host, the orientation of the worm, the orientation of the pygmy seahorse and the distance approximation between worm and pygmy seahorse.

surface without fully exiting it (figure 3), further indicating their reliance on the gorgonian for protection, combined with a need to frequently interact with the exterior. Some iNaturalist photographs showed a worm's anterior body going from a polyp burrow towards a branch burrow (iNaturalist IDs 148081426, 38953421), suggesting they may use it to quickly switch galleries. Based on the observation of posterior parts of worms protruding out of burrows, a situation that we have also observed in our dataset (iNaturalist ID 92708796), Martin et al. [29] suggested that worms temporarily exit galleries. However, we did not observe *H. anthogorgicola* completely exited from burrows or galleries, except for occurrences on the pygmy seahorse itself (see below discussion).

## (d) Interactions between *Haplosyllis anthogorgicola*, *Hippocampus bargibanti* and Anthogorgiidae

Both *H. bargibanti* and *H. anthogorgicola* inhabit anthogorgiid hosts, and were consistently observed in the same colony, suggesting they are likely to interact, potentially in a way that is significant for their survival. The most straightforward interaction to consider would be that *H. bargibanti* feeds on the worms, which could be a viable food source given their high abundance in the coral. Indeed, tripartite relationships between *Haplosyllis* worms and fish sharing the same host have been previously suggested [23,54], with *Haplosyllis* being the main prey found in the gut content of several sponge-associated gobies [55]. The exact diet of *H. bargibanti* in the wild has not been documented, but it has been observed feeding from open polyps or adjacent areas (instead of filtering plankton) in laboratory conditions [17]. This behaviour was interpreted as them feeding on food particles caught by the polyp. Alternatively, we suggest that the 'near-polyp' feeding behaviour of the seahorse may imply eating the worms. In some photographs (IDs 38196264, 148996869), the pygmy seahorse mouths can be seen in very close contact with worms. However, more direct evidence is needed to confirm the hypothesis of a trophic relationship, as seahorse species have been shown to feed mainly on crustaceans or nematodes, with polychaetes found in low abundances in their gut content [56–59]. Although *H. bargibanti* is in danger of extinction [33], attempts at rearing the seahorses have faced difficulty, mostly owing to insufficient food intake of the seahorses [17] and also owing to their gorgonian hosts, which as azooxanthellate corals are considered more challenging to rear [60,61]. A better understanding of the potential role of *H. anthogorgicola* in the pygmy seahorse's diet or host persistence could contribute to future endeavours into the rearing of *H. bargibanti*, and the conservation of this species.

Alongside or beside a trophic interaction, other relationships between the worm, the pygmy seahorse and the gorgonian can be hypothesized. Symbiotic polychaetes may show both intraspecific [62,63] and interspecific [64] antagonistic behaviours. In the case of sponge-associated *Haplosyllis*, some reports suggest that they may be protecting their hosts from attacks by predatory fish and sea slugs by intimidating or biting them. In turn, fish were observed biting the sponges, prompting the worms to protrude out of their hosts as a defensive behaviour, at which point they were preyed upon by the fish [23]. A similar interaction was suggested

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for *Haplosyllis basticola* Sardá, Avila & Paul, 2002, which inhabits the sponge *Ianthella basta* (Pallas, 1766), while the goby *Pleurosycia elongata* Larson, 1990 lives on the surface of the sponge [54]. The hypothetical host-guarding behaviour of symbiotic worms has been discussed in relation to photographs shared on online forums [65], which show syllid worms inhabiting sponges (likely *Haplosyllis*) protruding out of their host, seemingly attacking close-by spongivorous nudibranchs (http://www.seaslugforum.net/find/17308; http://www.seaslugforum.net/showall/hypsfest). A similar interaction may be considered for *H. anthogorgicola*, *H. bargibanti* and the anthogorgiid hosts. Threat displays are behaviour during which an animal makes itself appear intimidating to fend off a predator or opponent [66]. In some iNaturalist records, the behaviour of worms may be interpreted as a threat display towards the pygmy seahorse (e.g. iNaturalist ID 92708796), where several worms are visibly protruding their anterior body, standing vertically and pointing towards the pygmy seahorse (worm 2 in figure 5), which may be interpreted as a display of intimidation. The proximity of a predator or opponent represents a risk that can trigger threat displays [67]. Our results show that worms orientated towards the pygmy seahorse were associated with the pygmy seahorse being at a distance allowing contact or being relatively close (figure 6), suggesting this may represent a threat display. Furthermore, worms completely outside the colony were only seen when crawling on the surface of a pygmy seahorse (iNaturalist record IDs 2104253, 154483654, 31750768; worm 1 in figure 5), likely being able to bite and further deter it. This may be considered as 'mobbing' as several worms showed this behaviour simultaneously, and this activity could also alert nearby conspecifics [68]. Although non-significant, our analyses show that worms tended to point at a pygmy seahorse when it was oriented towards them (figure 6). These apparent aggressive behaviours may aim at preventing the seahorse either from eating them or from stealing food from the coral. However, considering the relatively small number of pygmy seahorses, it is unclear whether any potential guarding activity of the worms against seahorses and/or other symbionts would be beneficial enough for the corals to counterbalance the load of hosting such a large worm population within their tissue. This may depend on the relationships between the worms and their host gorgonians, which warrant further research.

## 5. Conclusions

Gorgonian hosts of *H. anthogorgicola* are typically found at 15–40 m depth in turbulent locations, making it difficult to observe them and their symbionts [69]. Recently, the value of citizen science photographs to overcome such challenges has been highlighted. In particular, photographs that include the surrounding environment of the species of interest have proven useful to maximize the information available about the targeted species and our understanding of its relationships with co-occurring organisms [53,70]. Accordingly, photographs of the pygmy seahorse including details of the host colony coincidentally helped investigate the presence of *H. anthogorgicola.* Although the species has not been recorded for nearly 70 years, pictures indicating the presence of *H. anthogorgicola* or a closely related species have been available on iNaturalist since 2012 (four years after the launch of the platform), and the worms' presence was detected in three-quarters of all photographs of the pygmy seahorse. These photographs allowed us to significantly extend the worm's putative distribution across the Indo-Pacific, as well as to describe its life habits and its interactions with the host gorgonian and pygmy seahorse. Our results show that these polychaetes occupy burrows throughout the whole gorgonian colony, most frequently in a position allowing them to be minimally exposed while sensing the surrounding environment, as well as partially exiting burrows to interact with elements on the host surface, hinting at a potential cleaning behaviour. Worms also appeared to show an antagonistic behaviour towards the pygmy seahorse that may correspond to defending themselves against being predated or defending host polyps from the seahorse stealing their food. This investigation shows that through abundant data available for a charismatic and popular species like *H. bargibanti*, citizen science databases may harbour valuable information on other overlooked, small and understudied co-symbionts such as *H. anthogorgicola*.

Ethics. This work did not require ethical approval from a human subject or animal welfare committee.

Data accessibility. Metadata of examined pygmy seahorse pictures, retrieved from iNaturalist, are provided in electronic supplementary material, table S1 [71]. Data on polychaete behaviour extracted from this dataset are provided in electronic supplementary material, table S2. Summary data on polychaete behaviours and their respective occurences is provided in electronic supllementary material, table S3.

Declaration of AI use. We have not used AI-assisted technologies in creating this article.

Authors' contributions. C.J.L.F.: conceptualization, data curation, formal analysis, investigation, methodology, software, validation, visualization, writing—original draft, writing—review and editing; L.M.: data curation, formal analysis, investigation, methodology, validation, writing—review and editing; J.A.A.L.: formal analysis, investigation, methodology, validation, visualization, writing—review and editing; A.T.: investigation, methodology, writing—review and editing; T.K.: data curation, formal analysis, investigation, methodology, resources, writing—review and editing; J.D.R.: funding acquisition, investigation, project administration, resources, supervision, validation, writing—review and editing. All authors gave final approval for publication and agreed to be held accountable for the work performed herein.

Conflict of interest declaration. We declare we have no competing interests.

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